

Laboratorio 03 - Detección de Ataque de Fuerza Bruta por SSH

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Rol objetivo : SOC Analyst Level 1

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Herramientas: Kali Linux, Ubuntu, VirtualBox

Objetivo

Simular un ataque de fuerza bruta mediante múltiples intentos fallidos de autenticación SSH y realizar el análisis de los registros del sistema para identificar evidencias del ataque, documentando el proceso como lo haría un analista SOC Nivel 1.

Alcance

Durante el laboratorio se analizaron los eventos generados por el servicio OpenSSH de Ubuntu, identificando:

- Intentos fallidos de autenticación.
 - Usuario objetivo.
 - Dirección IP origen.
 - Cantidad de intentos.
 - Evidencias del ataque.
-

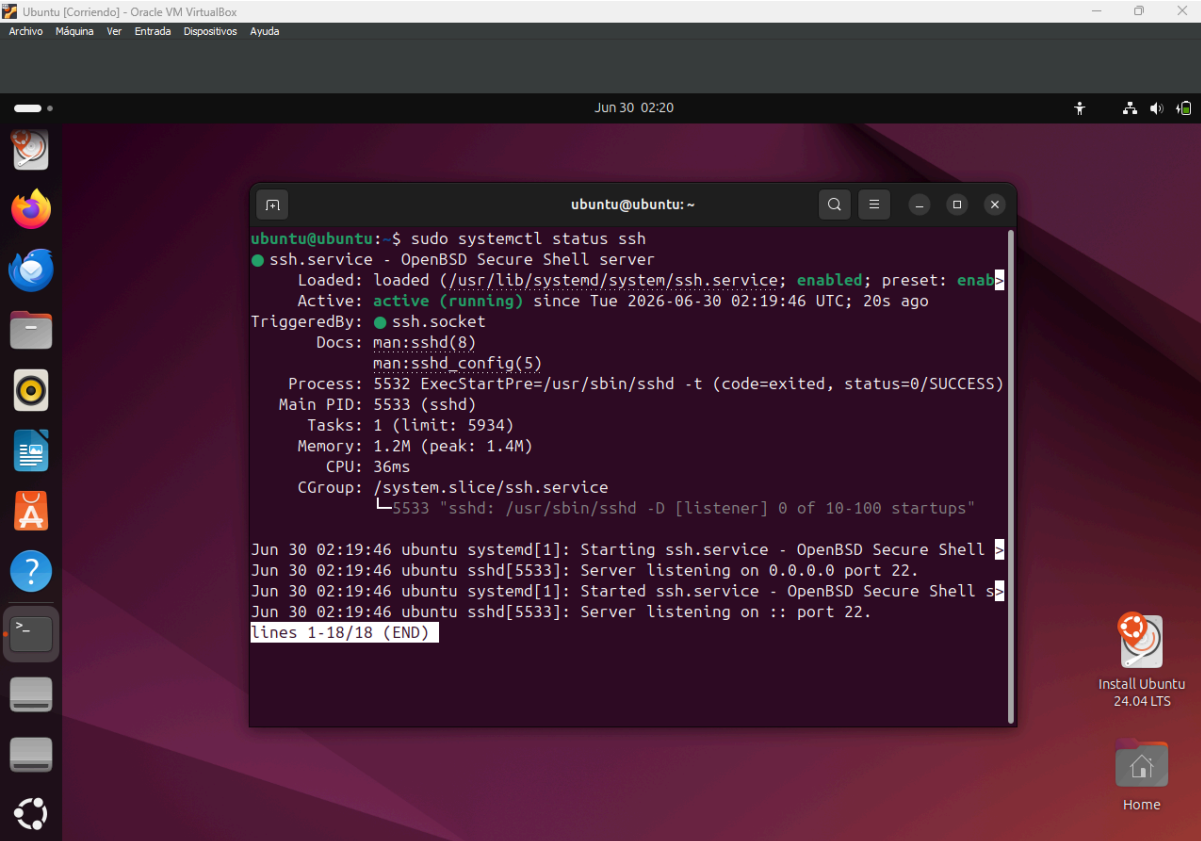
Entorno

Equipo	Sistema
Atacante	Kali Linux
Víctima	Ubuntu
Servicio	OpenSSH
Virtualización	VirtualBox

Desarrollo

Verificación del servicio SSH

Se verificó que el servicio OpenSSH estuviera activo y listo para aceptar conexiones.



The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~' with the following output from the command 'sudo systemctl status ssh':

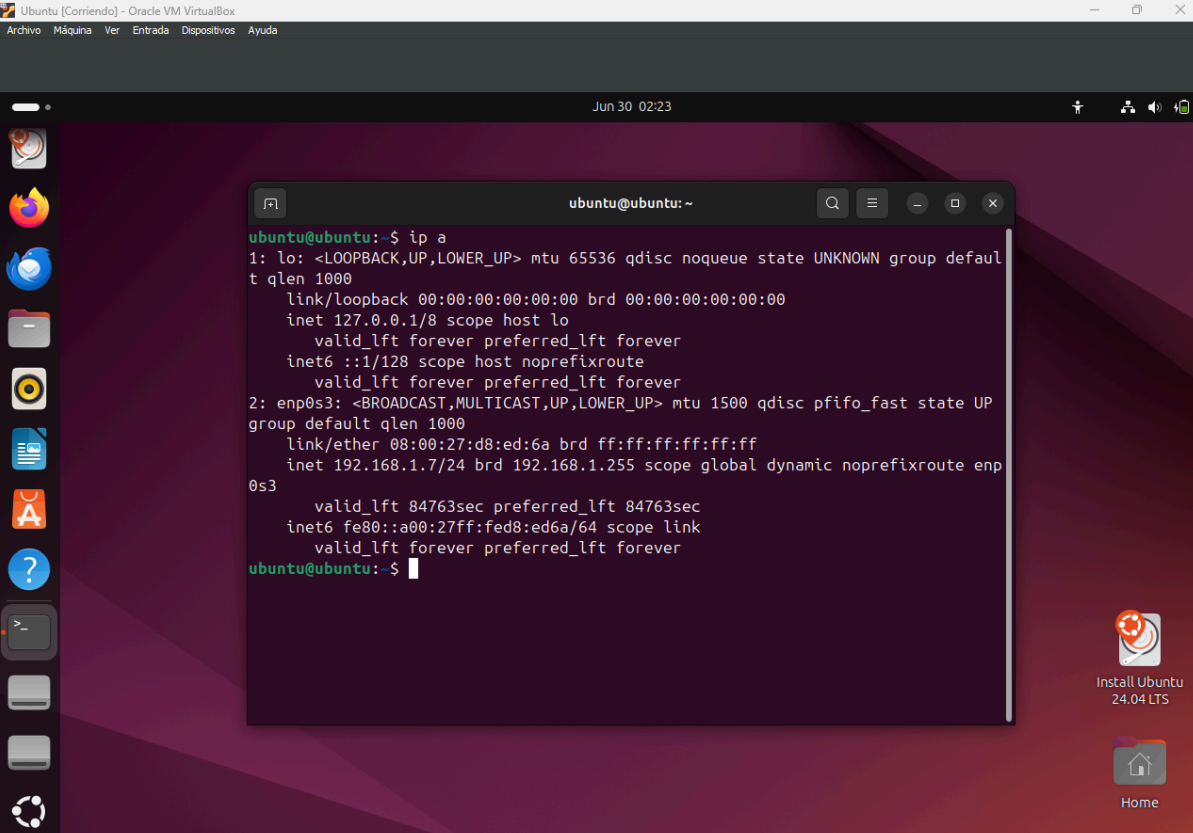
```
ubuntu@ubuntu:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-06-30 02:19:46 UTC; 20s ago
     TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 5532 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 5533 (sshd)
    Tasks: 1 (limit: 5934)
   Memory: 1.2M (peak: 1.4M)
      CPU: 36ms
   CGroup: /system.slice/ssh.service
           └─5533 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 30 02:19:46 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell
Jun 30 02:19:46 ubuntu sshd[5533]: Server listening on 0.0.0.0 port 22.
Jun 30 02:19:46 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell
Jun 30 02:19:46 ubuntu sshd[5533]: Server listening on :: port 22.
lines 1-18/18 (END)
```

The terminal window is open on a desktop environment with a dark purple background. The desktop has a sidebar on the left with various application icons and a top bar showing the date and time as 'Jun 30 02:20'. In the bottom right corner, there are icons for 'Install Ubuntu 24.04 LTS' and 'Home'.

Obtención de la dirección IP del servidor.

Se identificó la dirección IP que posteriormente sería utilizada por el equipo atacante.

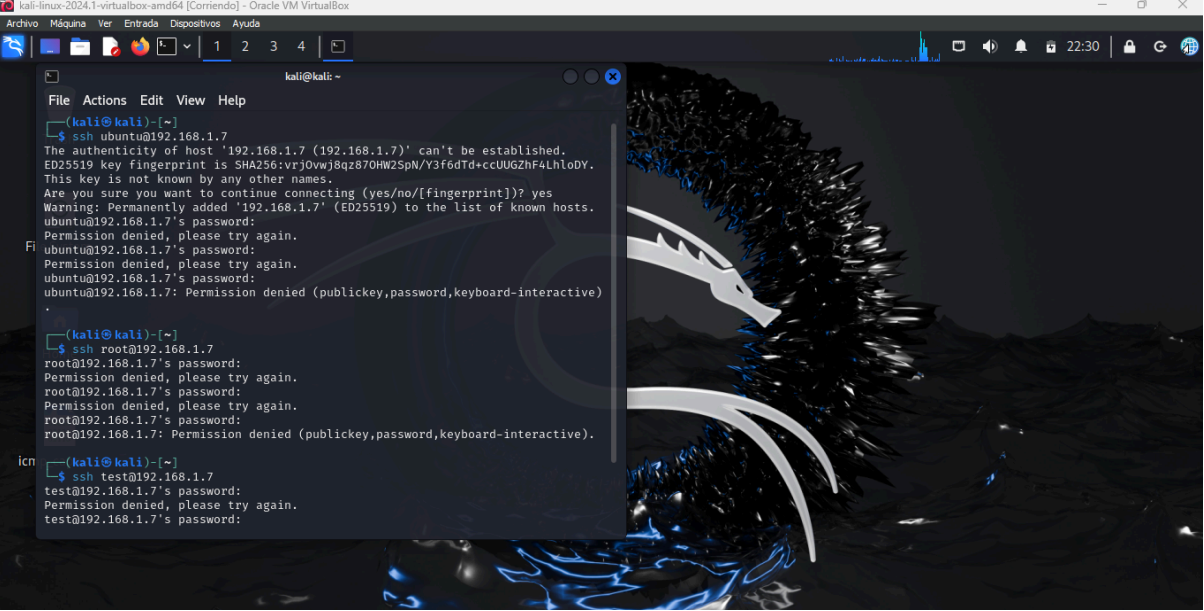


The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~' within an Oracle VM VirtualBox environment. The terminal displays the output of the command 'ip a'. The output shows details for the loopback interface 'lo' (127.0.0.1) and the ethernet interface 'enp0s3' (192.168.1.7). The terminal window is overlaid on the Ubuntu desktop, which includes a sidebar with application icons and desktop icons for 'Install Ubuntu 24.04 LTS' and 'Home'.

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
    group default qlen 1000
    link/ether 08:00:27:d8:ed:6a brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.7/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 84763sec preferred_lft 84763sec
    inet6 fe80::a00:27ff:fed8:ed6a/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

Generación del Ataque

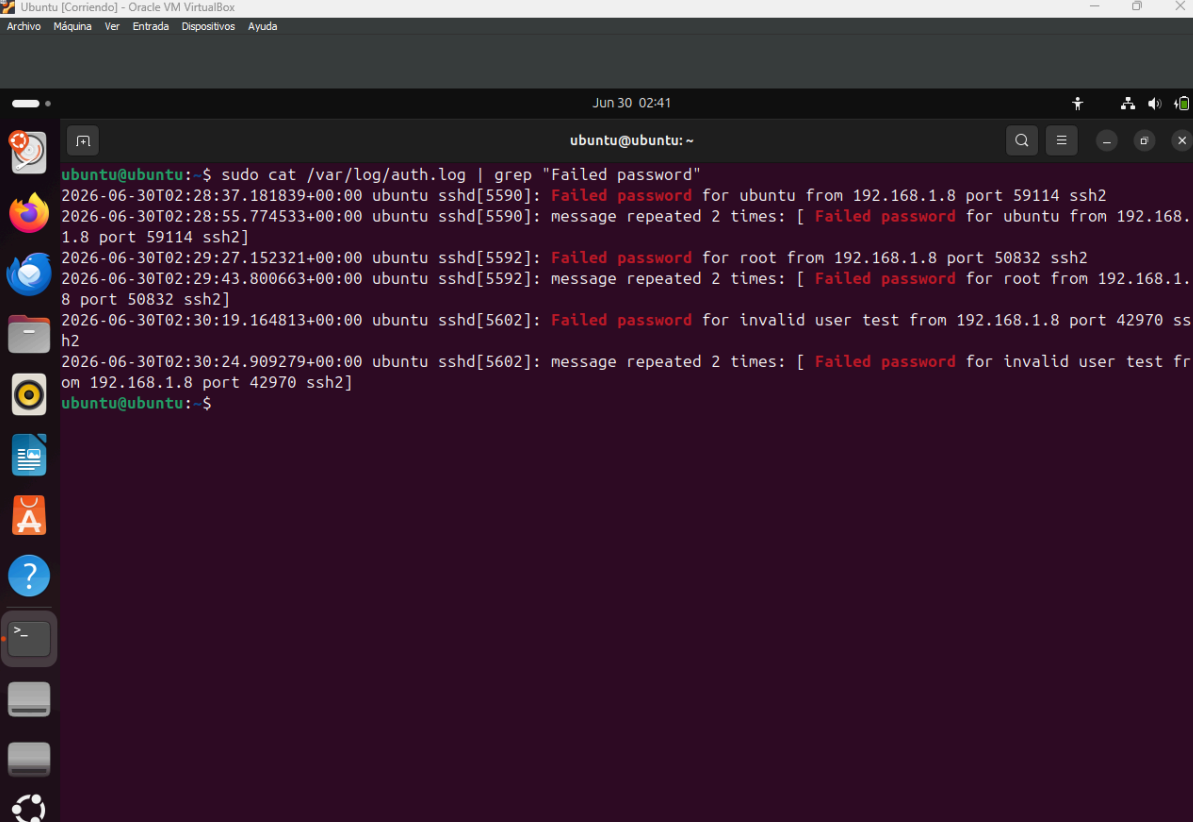
Desde Kali Linux se realizaron múltiples intentos de autenticación utilizando diferentes usuarios y contraseñas incorrectas.



```
kali@kali: ~  
File Actions Edit View Help  
❯ ssh ubuntu@192.168.1.7  
The authenticity of host '192.168.1.7 (192.168.1.7)' can't be established.  
ED25519 key fingerprint is SHA256:vrj0Vwj8qz870HW2SpN/Y3f6dTd+ccUUGZhF4LhLoDY.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.1.7' (ED25519) to the list of known hosts.  
ubuntu@192.168.1.7's password:  
Permission denied, please try again.  
ubuntu@192.168.1.7's password:  
Permission denied, please try again.  
ubuntu@192.168.1.7's password:  
ubuntu@192.168.1.7: Permission denied (publickey,password,keyboard-interactive).  
  
❯ ssh root@192.168.1.7  
root@192.168.1.7's password:  
Permission denied, please try again.  
root@192.168.1.7's password:  
Permission denied, please try again.  
root@192.168.1.7's password:  
root@192.168.1.7: Permission denied (publickey,password,keyboard-interactive).  
  
❯ ssh test@192.168.1.7  
test@192.168.1.7's password:  
Permission denied, please try again.  
test@192.168.1.7's password:
```

Revisión de Registros

Los registros del archivo `/var/log/auth.log` permitieron observar todos los intentos fallidos.

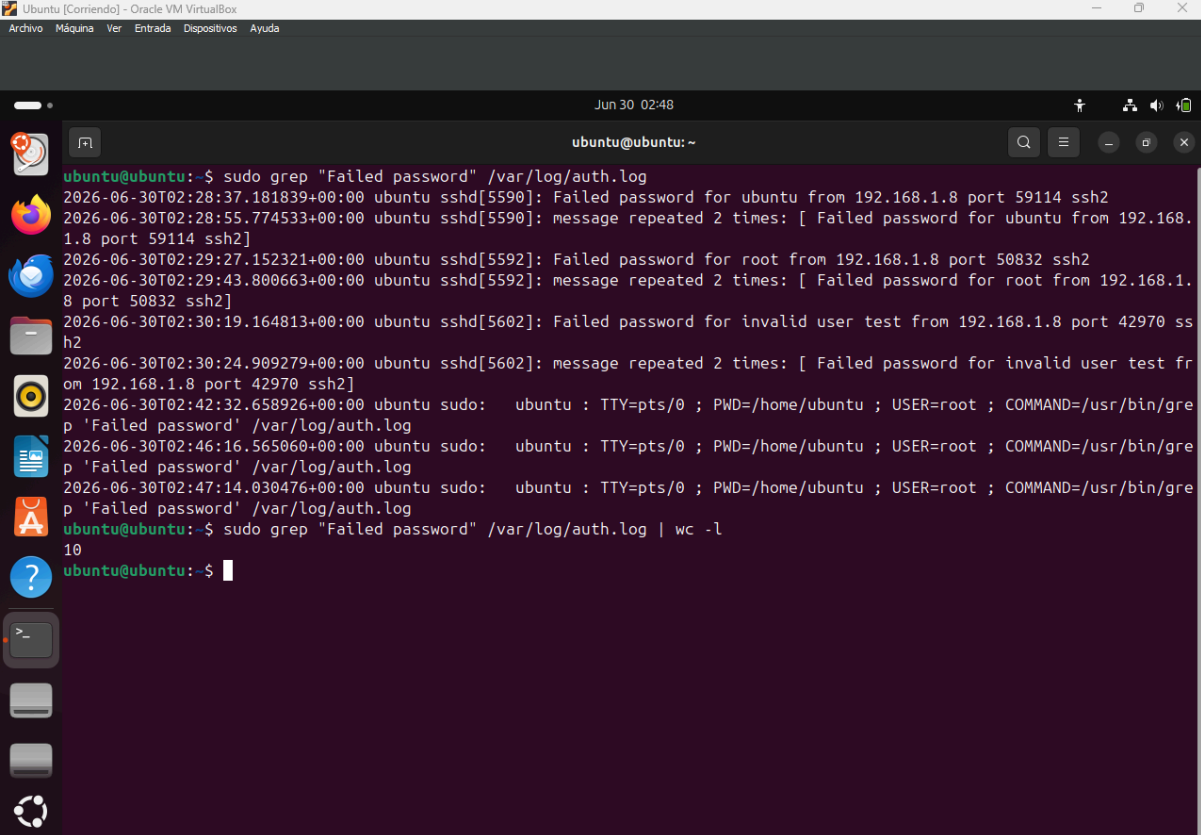


The screenshot shows a terminal window titled 'Ubuntu [Corriendo] - Oracle VM VirtualBox'. The terminal output displays the command `sudo cat /var/log/auth.log | grep "Failed password"` and its results. The results show several failed password attempts for the 'ubuntu' and 'root' users from the IP address 192.168.1.8. The messages are repeated twice for each attempt.

```
ubuntu@ubuntu:~$ sudo cat /var/log/auth.log | grep "Failed password"
2026-06-30T02:28:37.181839+00:00 ubuntu sshd[5590]: Failed password for ubuntu from 192.168.1.8 port 59114 ssh2
2026-06-30T02:28:55.774533+00:00 ubuntu sshd[5590]: message repeated 2 times: [ Failed password for ubuntu from 192.168.1.8 port 59114 ssh2]
2026-06-30T02:29:27.152321+00:00 ubuntu sshd[5592]: Failed password for root from 192.168.1.8 port 50832 ssh2
2026-06-30T02:29:43.800663+00:00 ubuntu sshd[5592]: message repeated 2 times: [ Failed password for root from 192.168.1.8 port 50832 ssh2]
2026-06-30T02:30:19.164813+00:00 ubuntu sshd[5602]: Failed password for invalid user test from 192.168.1.8 port 42970 ssh2
2026-06-30T02:30:24.909279+00:00 ubuntu sshd[5602]: message repeated 2 times: [ Failed password for invalid user test from 192.168.1.8 port 42970 ssh2]
ubuntu@ubuntu:~$
```

Conteo de Eventos

Se contabilizó el número de intentos registrados para medir la intensidad del ataque.



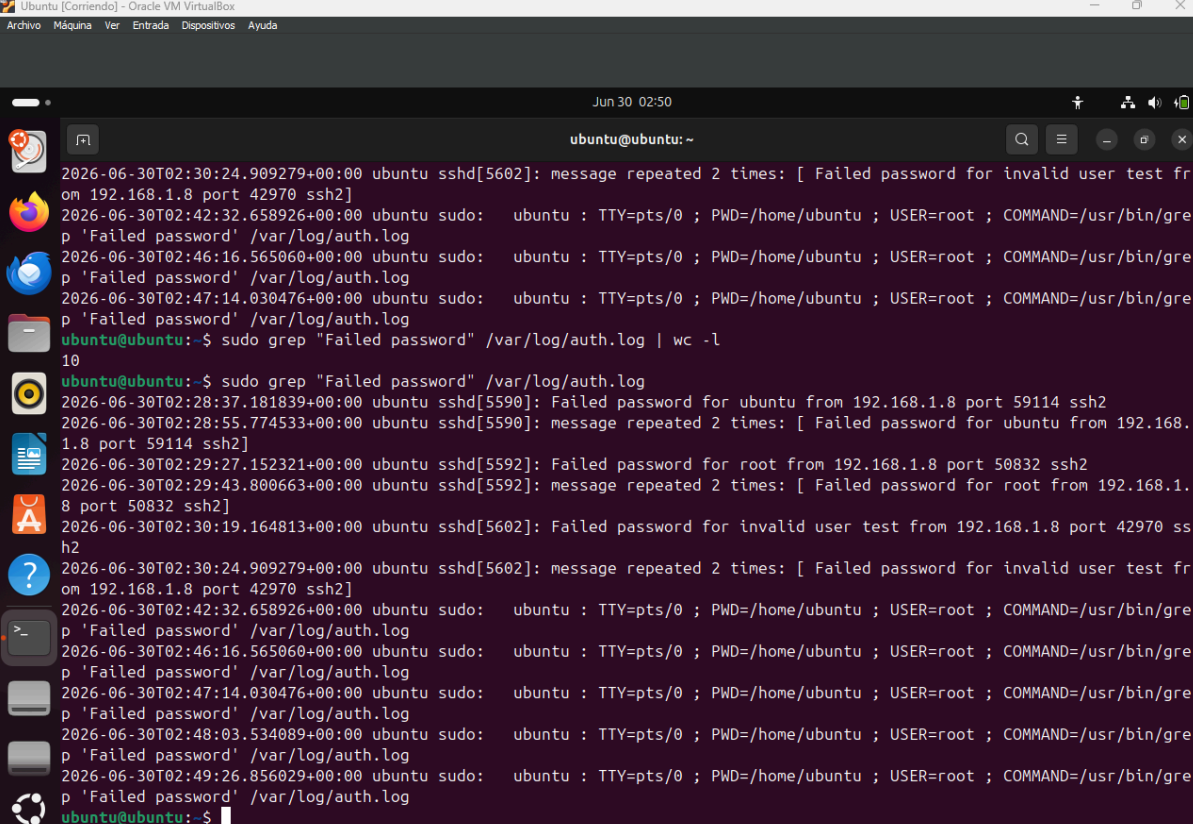
```
Ubuntu [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

Jun 30 02:48
ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log
2026-06-30T02:28:37.181839+00:00 ubuntu sshd[5590]: Failed password for ubuntu from 192.168.1.8 port 59114 ssh2
2026-06-30T02:28:55.774533+00:00 ubuntu sshd[5590]: message repeated 2 times: [ Failed password for ubuntu from 192.168.1.8 port 59114 ssh2]
2026-06-30T02:29:27.152321+00:00 ubuntu sshd[5592]: Failed password for root from 192.168.1.8 port 50832 ssh2
2026-06-30T02:29:43.800663+00:00 ubuntu sshd[5592]: message repeated 2 times: [ Failed password for root from 192.168.1.8 port 50832 ssh2]
2026-06-30T02:30:19.164813+00:00 ubuntu sshd[5602]: Failed password for invalid user test from 192.168.1.8 port 42970 ssh2
2026-06-30T02:30:24.909279+00:00 ubuntu sshd[5602]: message repeated 2 times: [ Failed password for invalid user test from 192.168.1.8 port 42970 ssh2]
2026-06-30T02:42:32.658926+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
2026-06-30T02:46:16.565060+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
2026-06-30T02:47:14.030476+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
ubuntu@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log | wc -l
10
ubuntu@ubuntu:~$
```

Identificación de la IP atacante

Se confirmó la dirección IP origen responsable de todos los intentos de autenticación.



```
Ubuntu [Corriendo] - Oracle VM VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

Jun 30 02:50
ubuntu@ubuntu: ~

2026-06-30T02:30:24.909279+00:00 ubuntu sshd[5602]: message repeated 2 times: [ Failed password for invalid user test fr
om 192.168.1.8 port 42970 ssh2]
2026-06-30T02:42:32.658926+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
2026-06-30T02:46:16.565060+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
2026-06-30T02:47:14.030476+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
ubuntu@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log | wc -l
10
ubuntu@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log
2026-06-30T02:28:37.181839+00:00 ubuntu sshd[5590]: Failed password for ubuntu from 192.168.1.8 port 59114 ssh2
2026-06-30T02:28:55.774533+00:00 ubuntu sshd[5590]: message repeated 2 times: [ Failed password for ubuntu from 192.168.
1.8 port 59114 ssh2]
2026-06-30T02:29:27.152321+00:00 ubuntu sshd[5592]: Failed password for root from 192.168.1.8 port 50832 ssh2
2026-06-30T02:29:43.800663+00:00 ubuntu sshd[5592]: message repeated 2 times: [ Failed password for root from 192.168.1.
8 port 50832 ssh2]
2026-06-30T02:30:19.164813+00:00 ubuntu sshd[5602]: Failed password for invalid user test from 192.168.1.8 port 42970 ss
h2
2026-06-30T02:30:24.909279+00:00 ubuntu sshd[5602]: message repeated 2 times: [ Failed password for invalid user test fr
om 192.168.1.8 port 42970 ssh2]
2026-06-30T02:42:32.658926+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
2026-06-30T02:46:16.565060+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
2026-06-30T02:47:14.030476+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
2026-06-30T02:48:03.534089+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
2026-06-30T02:49:26.856029+00:00 ubuntu sudo:    ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/gre
p 'Failed password' /var/log/auth.log
ubuntu@ubuntu:~$
```

Hallazgos

- Se detectaron múltiples eventos **Failed password**.
- Los intentos estuvieron dirigidos a cuentas privilegiadas.
- Todos los eventos provinieron de una única dirección IP.
- El comportamiento observado coincide con un ataque de fuerza bruta mediante SSH.

Indicadores de Compromiso

- Failed password
 - Invalid user
 - Repeated login attempts
 - Source IP
 - SSH authentication failures
-

Recomendaciones

- Implementar autenticación mediante claves SSH.
 - Deshabilitar el acceso remoto para el usuario root.
 - Configurar Fail2Ban.
 - Aplicar políticas robustas de contraseñas.
 - Monitorear continuamente los registros de autenticación.
 - Configurar alertas para intentos consecutivos de autenticación.
-

Conclusiones

Durante el laboratorio fue posible generar y detectar exitosamente un escenario representativo de un ataque de fuerza bruta por SSH. El análisis de los registros permitió identificar el origen de los intentos de autenticación, el número de eventos y los usuarios objetivo, demostrando la importancia del monitoreo de los registros del sistema para detectar actividades maliciosas de forma temprana.

El laboratorio fortalece las competencias fundamentales de un Analista SOC Nivel 1, especialmente en la investigación de incidentes, análisis de evidencias y documentación técnica.